

EDUCATION

Johns Hopkins School of Medicine

Ph.D. candidate, Neuroscience, GPA: 4.00/4.00

- Kavli NDI Distinguished Graduate Fellow (2022 – 2024)

Baltimore, MD

2019 – Present

McGill University, Faculty of Science

B.Sc. Physiology First Class Honors, GPA: 3.98/4.00

Montreal, QC

2014 – 2018

RESEARCH EXPERIENCE

Johns Hopkins School of Medicine

PhD Candidate with Dr. Dwight Bergles

Thesis: Defining the rules governing myelin patterns and plasticity

- Designed a whole-brain imaging and analysis pipeline to visualize myelin patterns across injury and aging with light-sheet microscopy.
- Developed deep learning tools for automated structural tracing/tracking of neurons and oligodendrocytes.

Baltimore, MD

2019 – Present

Graduate Researcher – extended collaboration with Dr. Richard Haganir

2020 – 2023

- Designed a computational super-resolution algorithm (XTC) to super-resolve SEP-GluA2 labelled neuronal synapses *in vivo*, enabling the tracking of synaptic plasticity in living animals.

McGill University

Research Assistant with Dr. Jack Antel

- Developed a high-throughput therapeutic screening pipeline, using deep learning algorithms and automated microscopy to identify compounds for treating multiple sclerosis.

Montreal, QC

2017 – 2019

University of British Columbia

NSERC-USRA Research Assistant with Dr. Urs Hafeli

- Developed radioactive antibodies as a therapeutic treatment for pancreatic adenocarcinomas.

Vancouver, BC

Summer 2015

PUBLICATIONS

First Author Peer-Reviewed Publications:

Xu YKT*, Graves AR*, Coste GI, Haganir RL, Charles AS, Sulam J. “Cross-modality supervised image restoration enables nanoscale tracking of synaptic plasticity in living mice”. *Nature Methods*. (2023). *Equal contribution.

Xu YKT, Call CL, Sulam J, Bergles DE. “Automated *in vivo* Tracking of Cortical Oligodendrocytes.” *Frontiers in Cellular Neuroscience*. (2021).

Xu YKT, Chitsaz D, Brown RA, Cui QL, Dabarno MA, Antel JP, and Kennedy TE. “Deep learning for high throughput quantification of oligodendrocyte ensheathment at single-cell resolution.” *Communications Biology*. (2019).

Co-authored Peer-Reviewed Publications:

Heo D, Kim AA, Neumann B, Doze VN, **Xu YKT**, Mironova YA, Slosberg J, Goff LA, Franklin RJM, Bergles DE. "Transcriptional profiles of murine oligodendrocyte precursor cells across the lifespan". *Nature Aging*. (2025).

Drake SS, Charabti M, Simas T, **Xu YKT**, Maes E, Shi S, Antel JP, Prat A, Morquette B, Fournier A. "3-Dimensional immunostaining and automated deep-learning based analysis of nerve degeneration". *International Journal of Molecular Sciences*. (2022).

Chanoumidou K, Hernández-Rodríguez B, Windener F, [18 others, including **Xu YKT**], Kuhlmann T. "One-Step reprogramming of human fibroblasts into oligodendrocyte-like cells by SOX10, OLIG2, and NKX6.2." *Stem Cell Reports*. (2021).

Starost L, Lindner M, Herold M, **Xu YKT**, [13 others], Kuhlmann T. "Extrinsic immune cell-derived, but not intrinsic oligodendroglial factors contribute to oligodendroglial differentiation block in multiple sclerosis." *Acta Neuropathologica*. (2020).

Mozafari S, Starost L, Blandine M, Beatriz G, **Xu YKT**, [10 others], Evercooren AB. "Multiple sclerosis IPS-derived oligodendroglia conserve their properties to functionally interact with axons and glia in vivo." *Science Advances*. (2020).

Cui QL, Lin YH, **Xu YKT**, Fernandes MGF, Rao VTS, Kennedy TE, Antel JP. "Effects of Biotin on survival, ensheathment, and ATP production by oligodendrocyte lineage cells in vitro." *PLOS ONE*. (2020).

Kremer D, Gruchot J, Weyers V, Oldemeier L, Göttle P, Healy L, Jang JH, **Xu YKT**, Volsko C, Dutta R, Trapp BP, Perron H, Hartung H, Küry P. "pHERV-W envelope protein fuels microglial cell-dependent damage of myelinated axons in multiple sclerosis." *Proceedings of the National Academy of Sciences*. (2019).

Publications under review:

Xu YKT, Bush A, Musheyev E, Kim AA, Zhang S, Eugenin von Bernhardt J, Sulam J, Bergles JE. "Brain-wide mapping of oligodendrocyte organization and oligodendrogenesis across the murine lifespan". In revision at *Cell*.

Call LC, Neely SA, Early JJ, James OG, Zoupi L, **Xu YKT**, Williams AC, Chandran S, Lyons DA Monk KR, Bergles DE. "Flexible ensheathment of axons optimizes myelination of complex CNS networks." In revision at *Nature*.

Mironova YA, Dang B, Heo D, **Xu YKT**, Hsu AY, Eugenin von Bernhardt J, Molina-Castro GC, Kim AA, Lin JP, Reich D, Bergles DE. "Myelin is repaired by cell autonomous differentiation of oligodendrocyte progenitors." In revision at *Science*.

Kim AA*, Heo D*, **Xu YKT**, Doze VN, Gutman-Wei AY, Goff LA, Kawaguchi R, Bergles DE. "Identification of differentiating oligodendroglia reveals transcriptional changes associated with lineage progression and brain-wide differences in oligodendrogenesis". *Equal contribution. In revision at *Nature Neuroscience*.

INVITED AND SELECTED TALKS

Xu YKT, Bush A, Musheyev E, et al. “Brain-wide mapping of oligodendrocyte organization, vulnerability, and lifelong plasticity”. OneNeuro student talk, Johns Hopkins University (2025).

Xu YKT, Bush A, Musheyev E, et al. “Defining the myelinome: a brain-wide map of oligodendrocyte distribution and myelin patterning in health and disease”. Myelin Gordon Research Conference (2024). Selected as most “stunning” talk at the conference.

Xu YKT, Bush A, Call CL, et al. “Mapping brain-wide myelin patterns”. Department of neuroscience annual retreat, Johns Hopkins University (2022). One of two graduate students invited to present.

Xu YKT, Bush A, Graves A, et al. “Myelin mapping and tracking: DeepCellTrace and Cross-Trained CARE (XTC)”. Dr. Bradley Hyman group meeting, Massachusetts General Hospital (2022). Invited speaker.

CONFERENCE PRESENTATIONS

Xu YKT, Bush A, Musheyev E, et al. “Brain-wide mapping of oligodendrocyte organization, vulnerability, and lifelong plasticity”. St. Jude’s National Graduate Student Symposium (2025). Selected as one of 30 out of 1800 applicants.

Xu YKT, Bush A, Musheyev E, et al. “Defining the myelinome: a brain-wide map of oligodendrocyte distribution and myelin patterning in health and disease”. Young Investigators’ Day research award, Johns Hopkins (2025).

Xu YKT, Bush A, Musheyev E, et al. “Defining the myelinome: a brain-wide map of oligodendrocyte distribution and myelin patterning in health and disease”. Myelin Gordon Research Conference and Seminar (2024).

Xu YKT, Chitsaz D, Cui QL, et al. “Adaptable heuristic and deep learning algorithms for in vitro and in vivo quantification of oligodendrocyte ensheathment.” International Progressive MS alliance conference (2019).

Xu YKT, Chitsaz D, Cui QL, et al. “Automated pipeline using heuristic and deep learning algorithms for high-throughput quantification of oligodendrocyte ensheathment.” Society for Neuroscience conference (2018).

TEACHING, MENTORSHIP, AND LEADERSHIP

Johns Hopkins School of Medicine

Baltimore, MD

Instructor, Hopkins Engineering Applications & Research Tutorials (HEART) Aug. – Nov. 2022

- Designed and taught a 10-lecture course: “*Transgenic, optical, and computational tools for brain mapping*,” that introduced students to the engineering tools used for whole brain analysis.
- Provided weekly lectures, interactive demos, and a lab tour for a class of 13 students.

Mentor, Life Design Lab 2021 - 2023

- Mentoring students in imaging informatics and computer vision research projects.

Teaching Assistant, Neuroscience and Cognition I Sept. – Dec. 2020

- Assisted in preparation of weekly lectures and transition to online teaching.
- Provided students with support through review lectures and office hours.

McGill University *Montreal, QC*
Producer and Production Manager 2013 – 2019

- Established a production company, *Two Gents of New West*, that supported aspiring artists in the performing arts through youth run shows, grant writing, and annual workshops/events.
- Produced *Heathers: the musical* as part of McGill Arts Undergraduate Theatre Society. Managed a cast/crew of 60 students.

AWARDS AND SCHOLARSHIPS

Hans J. Prochaska research award, Young Investigator's Day, Johns Hopkins University 2025
Kavli NDI Distinguished Graduate Student Fellowship, Johns Hopkins University 2022 – 2024

- *Awarded \$80,000 over two years to support interdisciplinary neuroscience research*

Boo Anderson Scholarship in Science, McGill University 2017
Emily Ross Crawford Scholarship for Academic excellence in Science, McGill University 2015
National Science Research Council Award (NSERC-USRA), University of British Columbia 2015
Sigma Chi Leadership Scholarship, McGill University 2014

REFERENCES

Prof. Dwight Bergles
Professor of Neuroscience
Johns Hopkins University
dbergles@jhmi.edu

Prof. Richard Hugarir
Director, Department of Neuroscience
Johns Hopkins School of Medicine
rhugarir@jhmi.edu

Prof. Jeremias Sulam
Assistant Professor
Biomedical Engineering
Johns Hopkins University
jsulam1@jhu.edu